

SUMMARY REPORT

Pose-Consistent Image Generation with Attribute Diversity: Approach, Evaluation, and Insights

Introduction:

1) Technical Problem

Generating realistic images conditioned on a given input pose is a challenging task in computer vision and generative modeling. This problem requires the generation of diverse, photorealistic images that preserve the input pose while varying other attributes, such as:

1. **Gender** (e.g., male, female),
2. **Skin tone** (e.g., light, dark),
3. **Clothing style** (e.g., casual, formal, sportswear), and
4. **Backgrounds** (e.g., indoor, outdoor, complex textures).

The objective is to achieve a delicate balance between: pose consistency, attribute diversity, and photorealism. This problem is complex due to the combination of pose, attributes, and realism, making it an essential benchmark for generative models.

2) Key Challenges

The primary challenges associated with this problem include:

1. **Pose Preservation:**
 - Ensuring that the generated images adhere closely to the given input pose map.
 - Mitigating distortions or misalignments in challenging poses.
2. **Attribute Variations:**
 - Introducing controlled variations in gender, skin tone, clothing, and background while maintaining the underlying pose.
 - Preventing mode collapse (generating images with limited diversity).
3. **Realism vs. Structure:**
 - Balancing photo-realism with pose consistency and attribute diversity.
 - Avoid artifacts, blurriness, or unnatural appearances in complex scenarios (e.g., dynamic backgrounds, intricate clothing patterns, deformed faces, unnatural anatomy).
4. **Evaluation Complexity:**
 - Measuring pose consistency, attribute diversity, and realism simultaneously is non-trivial.
 - Existing metrics often fail to capture nuanced trade-offs between these factors.

3) Relevance

The ability to generate pose-consistent and diverse images has wide-ranging applications in various domains, such as Fashion and Virtual Try-Ons, Diversity Modeling, Human Motion Synthesis, Healthcare and Physical Therapy, etc This problem also serves as a benchmark for generative models, helping researchers evaluate their performance in preserving structure while introducing meaningful variations.

Solution and Methodology:

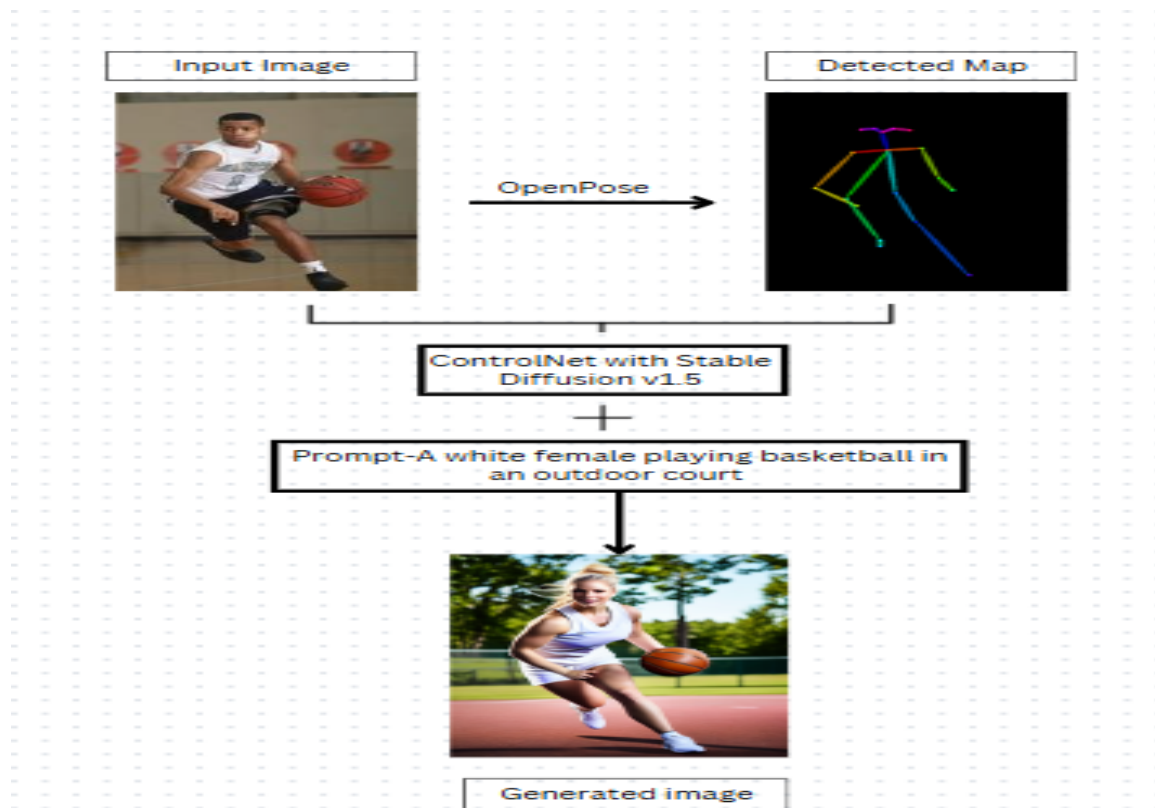
1) Approach Overview:

To tackle the problem of generating pose-consistent and diverse images, the solution leverages a **ControlNet architecture** conditioned on **Stable Diffusion v1.5** with **OpenPose keypoint maps** as the structural input. The pipeline incorporates controlled text prompts to guide the diversity of generated images across attributes like gender, skin tone, clothing, and backgrounds.

2) Pipeline:

- Input Pose Extraction:
 - The input image is processed through OpenPose, a state-of-the-art pose estimation model, to extract 2D key points representing the human pose.
 - The key points are visualized as a pose map, which serves as the structural guide for image generation.
- Text Prompt Conditioning:
 - To introduce diversity, carefully designed text prompts are used to control attributes such as:
 - Gender: "A black male in athletic clothes."
 - Clothing Style: "A female in casual wear."
 - Background: "Nature with trees" or "Indoor office setting."
 - These prompts are combined with the pose map to condition the ControlNet model.
- ControlNet with Stable Diffusion 1.5:
 - The ControlNet model conditions the latent diffusion process on the extracted pose map while using the prompt to steer the style and diversity of the output.
 - Stable Diffusion v1.5 acts as the base generative model, trained on large-scale datasets for photorealistic image synthesis.
- Image Generation:
 - The output is a collection of images that:
 - Maintain the pose from the input image.
 - Exhibit the desired variations in attributes (gender, skin tone, clothing, and background).
- Post-Processing :
 - Generated images are inspected for artifacts or inconsistencies.
 - If required, manual curation or additional prompts are used to refine the outputs.

3) Pipeline Diagram:



4) Post-Processing:

1) Fine-tuning prompts:

A key aspect of generating high-quality images involved ensuring that the model accurately interpreted the prompt and met the expectations for the output. Several challenges arose during generation, including:

- Anatomical Distortions: Unnatural body proportions or misaligned poses.
- Facial Deformities: Unrealistic or distorted facial features.
- Unrealistic Backgrounds: Surreal or unintended environmental settings.
- Clothing Artifacts: Inconsistent or unnatural clothing structures.

To address these issues, a significant portion of the post-processing was dedicated to fine-tuning prompts. This involved:

1. Adding Specific Details: Enhancing the clarity and specificity of prompts to align them with the desired output.
2. Negative Prompts: Including explicit instructions to avoid undesired features, such as "no distorted faces" or "no unrealistic lighting."
3. Weight Adjustments: Experiment with the weights assigned to prompts and negative prompts.
4. Control Strength Modifications: Adjusting the control strengths of the pose map within ControlNet to balance pose consistency with photorealism.

This iterative approach greatly reduced artifacts and improved the realism and quality of the generated images.

2) Edge Cases/Outliers of Pose Estimation:

Certain input images presented challenges where OpenPose failed to accurately detect 2D key points, leading to:

- Extreme Artifacts: Distorted or unrealistic images due to misidentified pose maps.
- Pose Loss: Failure to maintain the input pose in generated outputs.

To resolve these edge cases:

1. Alternative Methods: MediaPipe was evaluated as an alternative to OpenPose; however, it also produced unreliable results for challenging poses.
2. Manual Annotation:
 - Outlier images were manually annotated using the OnusAI annotation tool to generate accurate key points.
 - The annotated key points were integrated into the pipeline, replacing OpenPose's 2D detections.

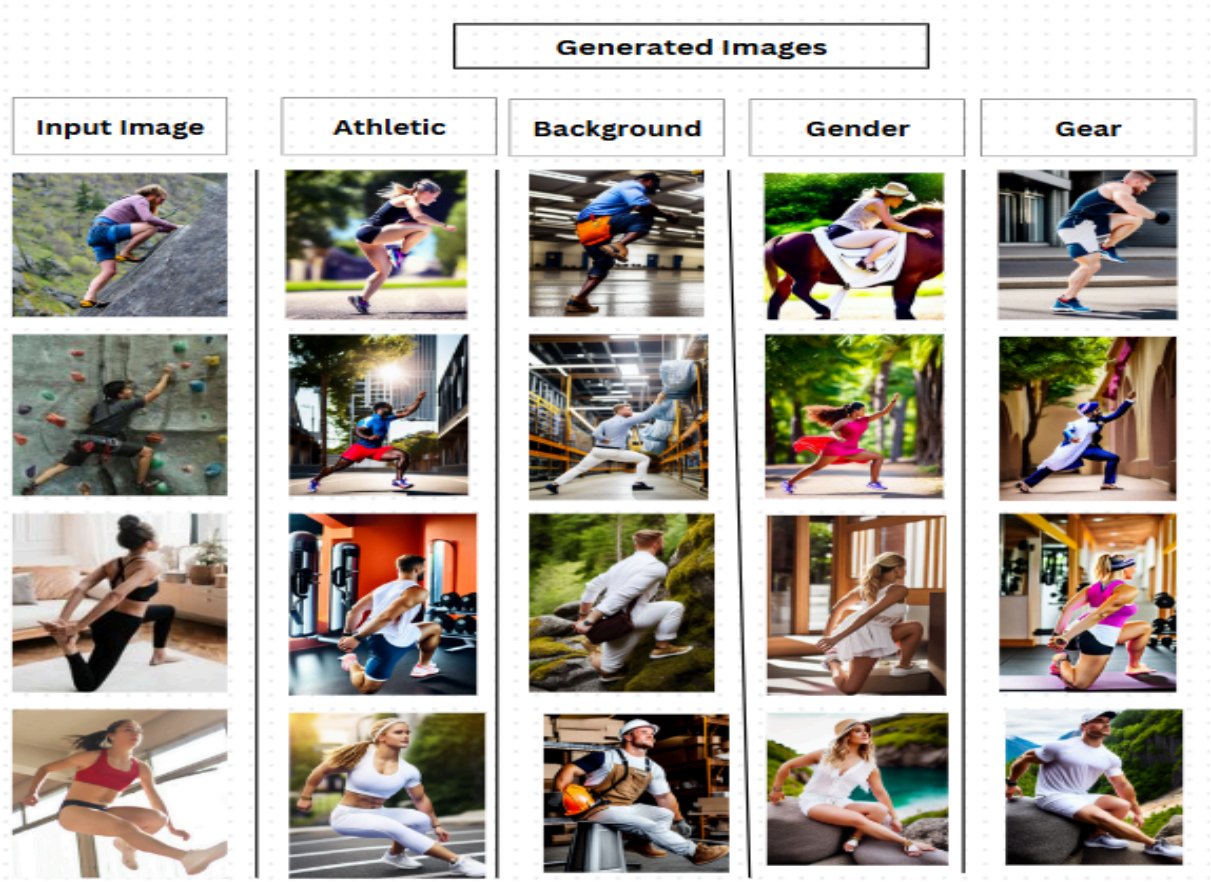
This solution significantly improved pose consistency and reduced artifacts, enabling ControlNet to produce higher-quality outputs even for challenging poses.

Before post-processing:	After post-processing:
 Pose Generated image	 Modified Pose Pose-consistent synthetic image

Results/Evaluation Metrics:

1) Qualitative Results:

A few results from the output set are included here:



Qualitative Analysis of Generated Images:

Aspect	Strength	Limitations
Pose Consistency	Accurate and well-preserved pose alignment across all outputs.	Rare distortions in extreme poses (e.g., climbing).
Attribute Variation	Strong diversity in gender, clothing, background, and props.	Slight blending artifacts in some accessories and clothing.
Photo-Realism	High-quality outputs with detailed textures and realistic lighting.	Minor artifacts in complex backgrounds or fine details.
Contextual Relevance	Outputs align well with prompts, creating contextually meaningful results.	Backgrounds can appear oversaturated or artificial in some cases.

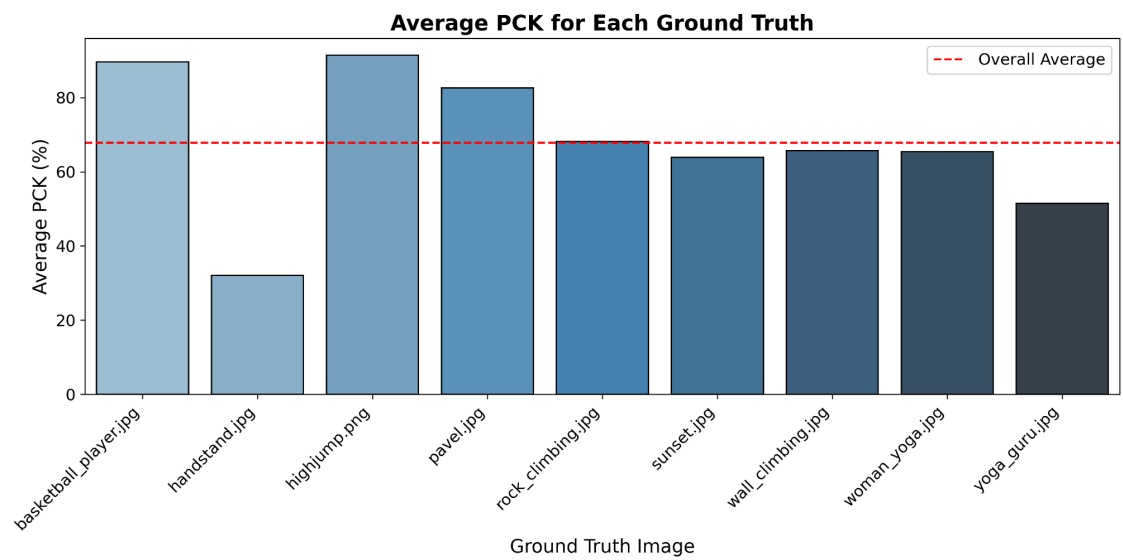
Evaluation Criteria:

Evaluation and Analysis

The evaluation of the model focused on three key aspects: pose consistency, photorealism, and image clarity. Metrics such as PCK (Percentage of Correct Key-points), SSIM (Structural Similarity Index), and Sharpness Score were used to quantify performance. Each metric was carefully selected for its relevance to the problem and provides a comprehensive assessment of the results.

1. Pose Consistency – Percentage of Correct Keypoints (PCK)

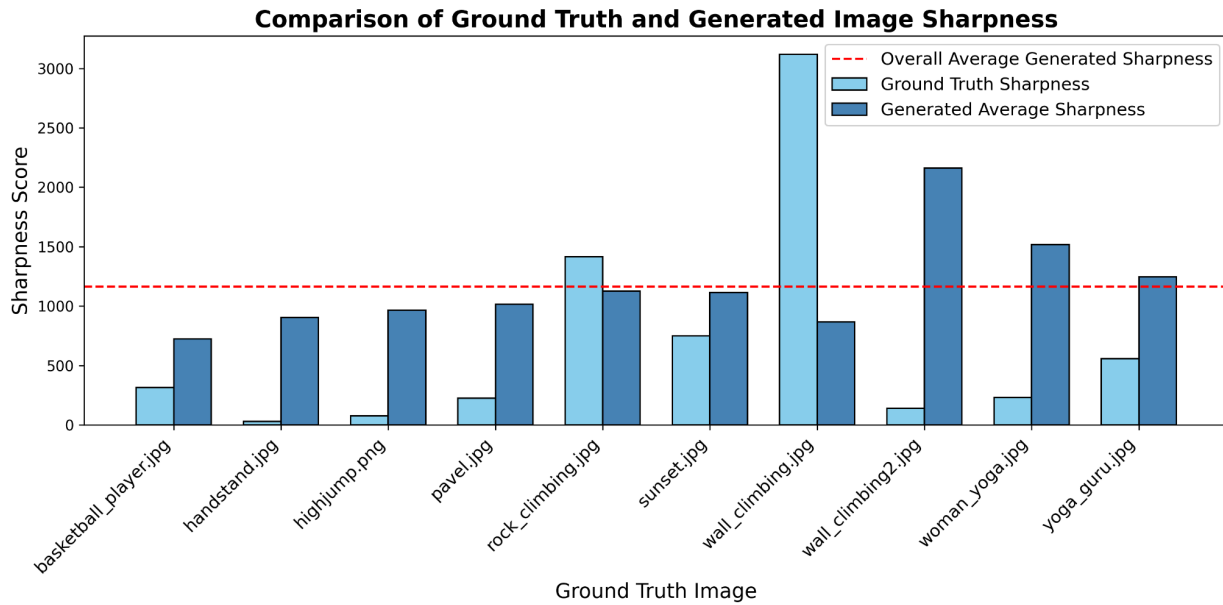
- **Why PCK?**
 - PCK was chosen to measure how accurately the generated images preserve the input pose. It evaluates the percentage of pose key points in the generated image that fall within a normalized threshold distance from the input pose key points.
 - This metric is highly relevant as the task requires pose consistency to be maintained, making PCK an essential quantitative measure.
- **Results:**
 - The average PCK score across all images was **67.83%**, indicating good overall pose consistency.
 - **Highest PCK Scores:**
 - **high jump (91.41%)** and **basketball_player (89.65%)** demonstrated strong pose preservation for static or well-aligned poses.
 - **Lowest PCK Scores:**
 - **handstand (32.12%)** and **wall climbing (65.66%)** highlighted challenges with dynamic or complex poses.
- **Insights:**
 - The model performs well for simpler poses but struggles with extreme articulation or occlusions.
 - **Future Improvement:** Incorporating 3D pose detection or depth maps could enhance consistency for challenging poses.



Ground Truth Image	Average PCK Score (%)	Analysis
Basketball player	89.65	Excellent pose consistency.
Handstand	32.12	Struggled with extreme articulation.
Highjump	91.41	Strong pose alignment for dynamic motion.
Pavel	82.64	Consistent pose preservation.
Rock_Climbing	68.18	Moderate success with complex poses.
Sunset	63.89	Average performance for scenic settings.
Wall_Climbing	65.66	Pose consistency impacted by occlusions.
Yoga_Woman	65.4	Retained most keypoints well.
Yoga_Male	51.52	Pose distortions in challenging postures.
Overall Average PCK	67.83	Decent overall pose consistency.

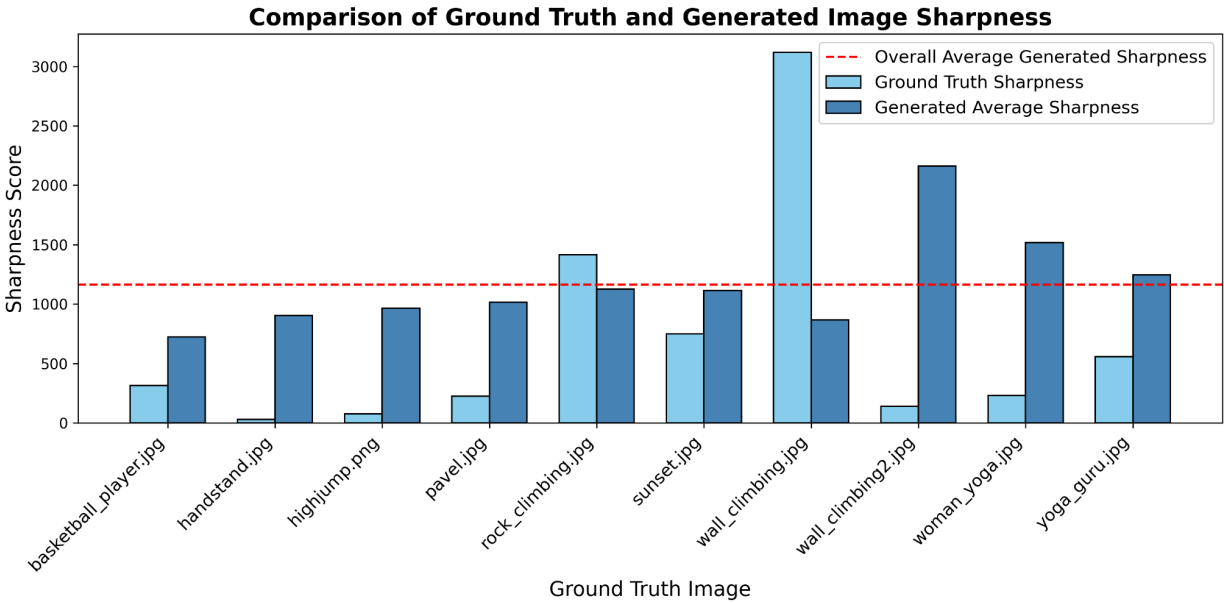
2. Photo-Realism – Structural Similarity Index (SSIM)

- **Why SSIM?**
 - SSIM measures perceptual similarity by comparing structural information, luminance, and contrast. This makes it suitable for evaluating how visually realistic the generated images are compared to the ground truth.
 - While intentional attribute diversity (e.g., changes in clothing or background) may reduce SSIM, the metric still ensures that structural alignment and visual quality are maintained.
- **Results:**
 - The average SSIM score was **0.329**, reflecting a pretty low structural similarity to the input.
 - **Highest SSIM Scores:**
 - **handstand (0.5736)** and **basketball_player (0.4812)** exhibited high visual alignment with the input pose and structure.
 - **Lowest SSIM Scores:**
 - **wall_climbing (0.1136)** and **rock_climbing (0.1457)** revealed challenges in maintaining structural alignment for visually complex scenes.
- **Insights:**
 - SSIM scores are lower for outputs with intentionally diverse attributes (e.g., backgrounds and clothing), as the metric penalizes deviations from the input image.
 - **Future Improvement:** Developing a custom metric that accounts for controlled diversity could better reflect the model's strengths.



3. Image Clarity – Sharpness Score

- **Why Sharpness?**
 - The sharpness score, calculated using the variance of the Laplacian, quantifies image clarity. Sharpness is particularly relevant for evaluating the visual appeal and detail retention in the generated outputs.
 - High sharpness scores indicate better clarity, while excessively high scores may suggest over-enhancement.
- **Results:**
 - The average sharpness score for generated images was **1163.56**, significantly higher than the ground truth in most cases.
 - **Highest Sharpness Scores:**
 - **wall_climbing (2162.08)** and **woman_yoga(1515.58)** showed improved sharpness, with textures and details well-preserved.
 - **Lowest Sharpness Scores:**
 - **wall_climbing (-2250.23)** and **rock_climbing (-289.31)** indicated artifacts or issues in texture rendering.
- **Insights:**
 - While higher sharpness enhances photo-realism, over-enhancement in some cases produces unnatural textures.
 - **Future Improvement:** Fine-tuning the model to balance sharpness and realism could mitigate these issues.



Ground Truth Image	Ground Truth Sharpness	Average Generated Sharpness	Difference
basketball_player.jpg	314.53	722.82	408.28
handstand.jpg	29.46	904.15	874.69
highjump.png	75.75	964.96	889.21
pavel.jpg	223.35	1015.46	792.12
rock_climbing.jpg	1415.11	1125.79	-289.31
sunset.jpg	750.73	1113.5	362.77
wall_climbing.jpg	3117.18	866.95	-2250.23
wall_climbing2.jpg	141.06	2162.08	2021.02
woman_yoga.jpg	229.47	1515.58	1286.11
yoga_guru.jpg	556.91	1244.27	687.37
Overall Average		1163.56	

Summary of Metric Relevance and Performance

Metric	Purpose	Why It Matters	Key Insights
PCK	Measures pose consistency.	Ensures generated images retain the input pose, which is critical for structural alignment.	Good overall pose consistency, but struggles with dynamic or occluded poses.
SSIM	Evaluate photo-realism and structural similarity.	Determines whether the generated images look realistic and align with the input visually.	Lower scores reflect intentional attribute diversity; future metrics could better capture this.
Sharpness	Assesses image clarity and detail.	Ensures outputs are sharp and visually appealing without artifacts or blur,	Improved sharpness, though over-enhancement leads to unnatural textures.

Fine-Tuning:

To enhance the model's ability to generate pose-consistent and attribute-diverse images, I implemented a fine-tuning process by creating a **custom dataset** tailored to the task requirements. This involved the following steps:

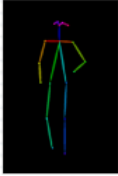
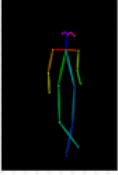
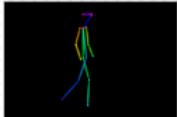
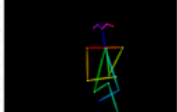
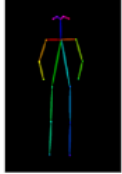
1. Dataset Curation

- **Web-Scraping for Images:**
 - I sourced images from various online repositories and web platforms, focusing on scenarios and subjects aligned with the desired attributes (e.g., gender diversity, clothing styles, and backgrounds).
 - Images were filtered to ensure relevance to the task and to minimize noise in the dataset.
- **2D Keypoint Annotation:**
 - OpenPose was used to extract 2D key points from the scraped images, generating pose maps for each sample in the dataset.
 - Manual annotation was performed for edge cases or poses where automated keypoint detection was unreliable.
- **Prompt Design:**
 - A set of carefully crafted prompts was created for each image, describing key attributes such as:
 - **Gender:** "A black female in sportswear."
 - **Clothing Style:** "A male in a suit in an office environment."
 - **Backgrounds:** "Outdoor forest setting" or "Urban industrial environment."
 - Negative prompts were included to explicitly suppress undesired artifacts (e.g., "no distorted faces," "no unrealistic lighting").

2. Fine-Tuning Process

- **Model Training:**
 - The curated dataset, consisting of pose maps, corresponding images, and attribute-specific prompts, was used to fine-tune the ControlNet model.
 - Fine-tuning was performed on a smaller subset of **Stable Diffusion 1.5** with ControlNet as the backbone, focusing on:
 - Preserving pose consistency.
 - Introducing robust diversity in attributes.
 - Enhancing photo-realism and reducing artifacts.
- **Training Configuration:**
 - The model was trained using a **learning rate schedule** optimized for fine-tuning tasks, ensuring convergence without overfitting.
 - Regularization techniques such as **data augmentation** (e.g., flipping, rotation, and brightness adjustments) were applied to improve generalization.

Part of the dataset structure:

Ground Truth	Conditional image(pose)	Prompt
		Fair skin female wearing athletic clothes and footwear
		Fair skin female wearing casual clothes and footwear indoor
		Fair skin male wearing sporty clothes and footwear
		Fair skin female wearing casual clothes and footwear indoor
		Dark skin male wearing casual clothes and footwear

Results

The fine-tuning process resulted in **poor performance**, with several notable issues:

- Pose Distortions:**
 - Generated images often failed to maintain pose consistency, particularly for challenging poses.
 - Misaligned key points or artifacts appeared in cases with dynamic or occluded poses.
- Reduced Photorealism:**
 - The outputs exhibited significant degradation in visual quality, including:
 - Blurry textures.
 - Unrealistic clothing and facial features.
 - Poorly rendered backgrounds.
- Inconsistent Attribute Representation:**
 - Despite carefully crafted prompts, the model struggled to reliably reflect the intended attributes (e.g., gender, clothing style, and environment).
 - Attribute diversity was limited, with outputs frequently collapsing into a narrow range of variations.

Possible Reasons for Poor Results

1. **Limited Dataset Size:**
 - The custom dataset was relatively small compared to the scale of pre-training datasets used for ControlNet and Stable Diffusion.
 - Insufficient diversity and volume likely caused overfitting to the fine-tuning dataset, leading to poor generalization.
2. **Data Quality Issues:**
 - Web-scraped images varied significantly in quality, with some containing noise, poor lighting, or irrelevant features.
 - Even with manual annotation, pose maps from edge cases were inconsistent, affecting the training signal.
3. **Inadequate Training Resources:**
 - Fine-tuning was performed with limited computational resources, potentially causing suboptimal convergence.
 - A more extensive training schedule or better hyperparameter tuning may have been required for meaningful improvements.
4. **Pre-Trained Model Constraints:**
 - ControlNet's pre-trained weights are heavily biased towards large-scale, diverse datasets. Fine-tuning on a much smaller and less diverse dataset may have disrupted these well-optimized parameters.

Insights:

1. **The Importance of Dataset Quality:**
 - High-quality, diverse data is essential for effective fine-tuning. The custom dataset lacked sufficient diversity and quality to rival the original pre-trained data.
2. **Scale Matters:**
 - Models like ControlNet and Stable Diffusion are pre-trained on millions of images. Fine-tuning requires a comparable scale or very carefully curated data to avoid overfitting and poor generalization.
3. **Prompt Engineering Alone is Insufficient:**
 - While prompts can influence outputs, they cannot compensate for fundamental issues in training data or pose maps.
4. **Future Directions:**
 - Use synthetic datasets generated from existing pre-trained models (like COCO or DeepFashion) to increase diversity and quality.
 - Incorporate 3D key points or depth maps to improve pose representation for dynamic or challenging poses.
 - Leverage better computational resources for fine-tuning, ensuring convergence and stability.

References-

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